Fake News Detector - Project Report

Introduction

With the increasing spread of misinformation online, fake news detection has become a critical task. This project addresses the challenge by using machine learning and NLP to automate the classification of news as fake or real. The model was trained on a labeled data-sets of news articles and deployed the project using Stream-lit to create a simple and interactive web interface .

Abstract:

The Fake News Detector project aims to classify news articles as either fake or real using Natural Language Processing (NLP) techniques. It leverages machine learning models to analyze textual content and identify deceptive information. The web-based interface built with Stream lit allows users to input any news article and instantly receive a prediction of its authenticity

Tools Used:

- **Python:** Core programming language for implementation.
- **Data handling and preprocessing**: Pandas ,Numpy
- **Scikit-learn:** machine learning
- **For vectorization :** TF-IDF
- **For text cleaning and tokenization:** NLTK
- **Joblib:** For model serialization and loading
- **Streamlit:** For building the user interface.

Steps Involved in Building the Project:

- **Data Collection:** A labeled dataset of real and fake news articles was collected from Kaggle.
- **Import the libraries**: import the necessary libraries to build the project
- **Data Preprocessing:** Performed text cleaning, tokenization, stopword removal
- **Text Vectorization**: Initializes a TF-IDF Vectorizer to convert text into numerical form
- **Splitting the Data**: Splits the dataset into training and testing sets using train_test_split() with 25% test data and a random seed of 42
- **Model Training:** Trained machine learning models (such as Logistic Regression) to
- classify news as fake or real.
- **Model Evaluation:** Evaluated the model using accuracy and confusion matrix metrics,classification report.
- **Model serialization and loading:** Save the trained TF-IDF vectorizer and model to disk using joblib.dump(vectorizer, 'vectorizer.jb') joblib.dump(model, 'model.jb'), these saved files are later used in a Streamlit app for real-time predictions.
- **Deployment:** Developed an interactive simple user interface using Streamlit app that allows users to input news text and get real-time predictions

Conclusion:

The Fake News Detector project demonstrates the effective application of Natural Language Processing and Machine Learning in combating the spread of misinformation By transforming raw text data into structured numerical features using TF-IDF vectorization, the system can accurately distinguish between real and fake news based on linguistic patterns and contextual cues. The use of Logistic Regression provides a balance of interpretability and accuracy, making the model reliable for practical use. The integration with Streamlit enables a userfriendly, interactive interface that allows individuals to validate news content quickly without requiring any technical knowledge. This feature bridges the gap between data science and real-world usability, promoting awareness and media literacy among users.